Paleomagnetism of the Jacobsville Formation and the Apparent Polar Path for the Interval −1100 to −670 m.y. for North America

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Chemical, thermal, and alternating field (af) treatments performed on 257 specimens (37 sites) of the Jacobsville Formation show that the magnetization was acquired gradually over a very long period of time. Three phases of magnetization can be recognized and sequentially separated chemically. Thermal (up to 675°C) and af (up to 290 mT) treatments are unable to distinguish and separate the first two phases. The results are consistent with a three-phase magnetization model previously proposed and again demonstrate the necessity of performing chemical leaching experiments in red bed studies. The early phase remanence isolated chemically indicates that the Jacobsville is of upper Keweenawan age (≃−1100 m.y.). Poles obtained from three areas, Sault Ste. Marie (SSM) (175°E, 12°N), Marquette (182°E, 03°S), and the Keweenaw Peninsula (KP) (184°E, 10°S), indicate that the sediments of SSM are older than those of KP. A detailed study (nine sites: mean pole at 190°E, 16°S) of a 5.3-m stratigraphic section at Jacobsville shows significant apparent polar motion and the occurrence of at least one field reversal. The trend of the pole movement relative to North America was then southward. Vector analyses show that during the acquisition of the intermediate and late phase remanences the apparent pole started to move northward; by the end of the magnetization process it had reached high latitude (307°E, 72°N). Explanation of the Jacobsville results and other high-latitude poles requires a major revision of the apparent polar path during the Hadrynian. The proposed path executes a loop into the southern hemisphere (−1100 to −950 m.y.) and then roughly follows a great circle path (Hadrynian Track) from 160°E near the equator up to the present north geographic pole and down to the equator at 340°E (−670 m.y.). The controversial Grenville poles could easily be incorporated into that loop, supporting the hypothesis that the Grenville Province has always been an integral part of Laurentia. The Hadrynian Track, with its large pole displacement (180° in 300 m.y.), is an extremely useful stratigraphic tool that can be used to great advantage for global correlations of late Precambrian rock units.

INTRODUCTION

For over a century, a great deal of work has been done to establish the stratigraphic sequence of the rock units of the Lake Superior region. The early work was spurred by the importance of rich copper deposits associated with the Keweenaw Peninsula rocks. With the aid of geophysical techniques, recent work has focused on the structural development and tectonic implications of the Helikian Keweenawan basin, which may form part of a much larger major Precambrian rift system [*Smith et al.*, 1966; *Goldich et al.*, 1966*a*, *b*; *Muehlberger et al.*, 1966; *Lidiak et al.*, 1966; *Halls and West*, 1971; *Baer et al.*, 1974; *Wynne-Edwards*, 1976].

Throughout all these investigations the stratigraphic position of one of the thickest and most extensive sedimentary units of the area, the Jacobsville Formation, has remained elusive. *Hamblin* [1958] listed 34 authors, each of whom suggested different age assignments from Triassic to late Precambrian. In the absence of fossils and lack of exposures showing its stratigraphic relation with older and younger rocks, most correlations between the Jacobsville and red beds of Paleozoic ages have been based upon lithologic similarities. *Hamblin* [1958] showed that at Grand Island the upper Cambrian Munising Formation overlies the Jacobsville and suggested a middle or lower Cambrian age for the latter. This has been questioned by several workers who have suggested ages from lower to upper Keweenawan [e.g., *Dubois*, 1962; *Babcock*, 1975].

The Jacobsville was sampled twice for paleomagnetic studies. The sampling of *Dubois* [1962] was limited: one site from the Keweenaw Peninsula (KP) and two from Sault Ste. Marie (SSM). The difference between the two poles (174°E, 17°S for KP and 164°E, 05°N for SSM) is 24°. Dubois concluded that the Jacobsville was upper Keweenawan in age. *French* [1976] collected 15 sites distributed on the Keweenaw Peninsula and along the south shore of Lake Superior. The site mean precisions (k) are low, apparently because two magnetizations could not be separated. The mean pole obtained from nine of those sites is 171°E, 05°N, 23° away from the Keweenaw Peninsula pole of *Dubois* [1962]. French suggested that the Jacobsville is of Eocambrian age. Therefore the paleomagnetic data, although they favor a Precambrian age as opposed to a Cambrian age, leave the Jacobsville undetermined within a 500-m.y. time span. It then appears that more detailed laboratory work performed on an extensive collection is required to determine more accurately the age of the Jacobsville.

GEOLOGY AND SAMPLING

The Jacobsville is exposed on the eastern half of the Keweenaw Peninsula and outcrops along the south shore of Lake Superior as far east as Sault Ste. Marie (Figure 1). The beds are nearly flat lying. Owing to the combined effects of the relief of the Precambrian surface upon which it was deposited and subsequent erosion, the thickness is extremely diverse. It varies from more than 600 m in some parts of the Keweenaw Peninsula to less than 5 m a few kilometers southeast of Marquette, indicating that the Jacobsville thickens greatly to the north

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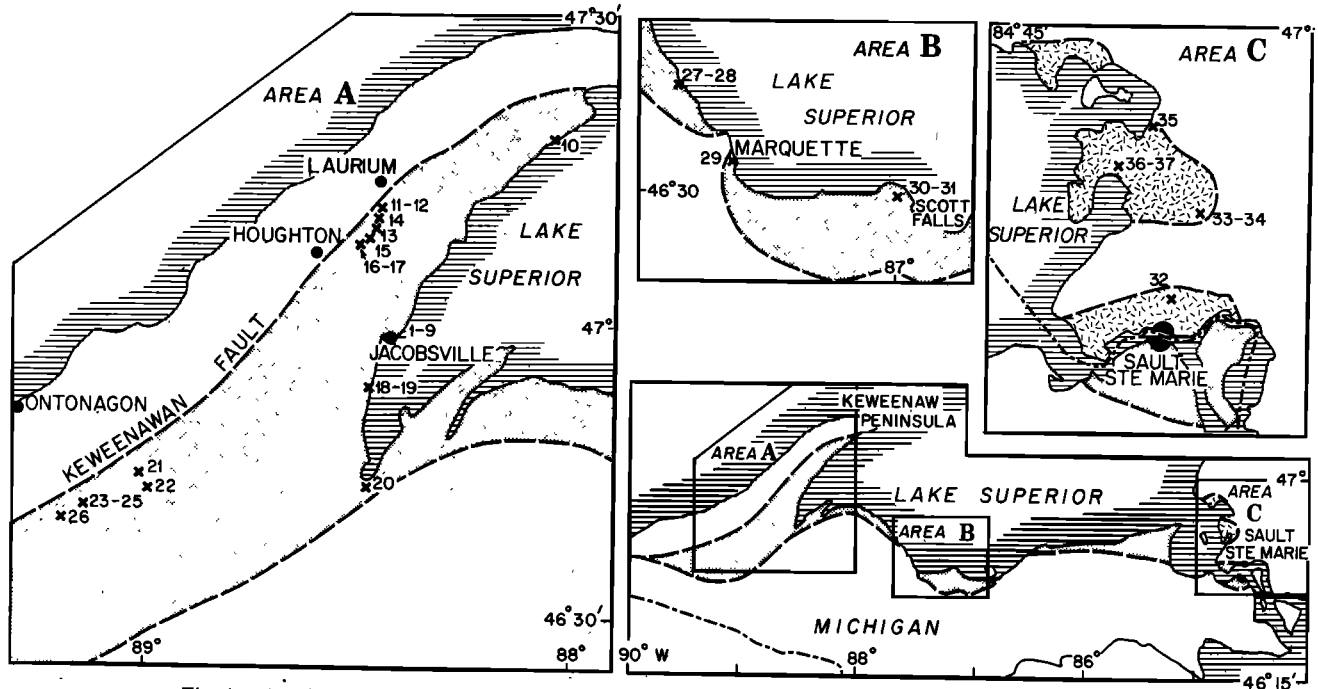


Fig. 1. Areal map of the Jacobsville Formation showing the sampling sites located in three areas.

[*Hamblin*, 1958]. This is in accordance with the seismic data of *Halls and West* [1971], who suggest that Jacobsville sandstones underlie most of Lake Superior.

The sandstones are extremely varied in both grain size and color. In the fine-grained units the basic red color is often mottled with white streaks or blotches. In the medium- and coarse-grained units the basic color may be red, brown, purple, beige, or pink and is often mottled with red or white streaks. According to *Hamblin* [1958] the mottling occurred after the coloring process and was produced by leaching.

Thirty-seven sites were sampled over a lateral distance of 300 km. Because of the wide range of lithologies and the variable thickness of the formation it is impossible to correlate them stratigraphically, and the results are divided into three parts defined by the areas A, B, and C (Figure 1). *Dubois* [1962] has suggested that the rocks at Sault Ste. Marie (area C) are older than those in the Keweenaw Peninsula (area A). *Babcock* [1975] has divided the Jacobsville sequence into upper, middle, and lower members, the upper member being found in the Keweenaw Peninsula.

Cleaning Methods

At each of the 37 sites, three to seven oriented core samples were drilled in the field; one or two standard size (2.5-cm diameter, 2.2-cm height) specimens were sliced from each core to yield 257 specimens, which were treated chemically, thermally, or by alternating field (af). The thermal and af experiments were performed in residual fields <2 nT [*Roy et al.*,

TABLE 1. Site Mean Results of the SA Section Obtained Before (NRM) and After Chemical, Thermal, or af Cleaning Treatment

	NRM				Treated*						
Site	Number of Specimens N	$D°, I°$†	Resultant Vector R	Precision Estimate k‡	Number of Specimens N_1	$D°, I°$†	Resultant Vector R	Precision Estimate k‡	Radius of Circle of Confidence α_{95}‡	Pole	Stratigraphic Separation,§ m
1	5	249, −18	4.805	21	5	249, −11	4.923	52	11	193°E, 18°S	
2	7	238, +00	6.913	69	7	238, −05	6.905	63	8	204°E, 23°S	1.0
3	7	256, +13	6.776	27	7	244, −01	6.899	59	8	261°E, 18°S	0.5
4	5	255, −11	4.966	118	5	252, −12	4.973	148	6	191°E, 17°S	0.1
5¶	6	268, +15	5.079	5	5	269, +03	4.722	15	21	186°E, 02°S	0.9
6	5	295, +08	4.915	47	5	269, −05	4.910	44	12	181°E, 03°S	0.9
7	5	264, −21	4.958	96	5	260, −21	4.966	116	7	181°E, 15°S	1.5
8	4	259, −23	3.922	38	4	254, −20	3.919	37	15	186°E, 19°S	0.3
9	3	358, −26	2.938	32	3	263, −18	2.974	78	14	180°E, 12°S	0.1
Mean	8	259, −10	7.498	14	8	254, −12	7.833	42	9	190°E, 16°S	

*These results are compiled (site 5 excluded) from 20 chemically treated, 19 thermally treated, and 2 af-treated specimens. In most instances, the treatment was 6700 hours, 675°C, or 290 mT.

†$D°$ and $I°$ are the declination east and inclination positive downward.

‡From *Fisher* [1953].

§Stratigraphic separations are between two adjacent sites, site 1 being the highest.

¶Not used for calculation of mean because of the occurrence of field reversal (see Figure 5); if it were used, the mean direction would be 255°, −10°, and the pole, 189°E, 14°S.

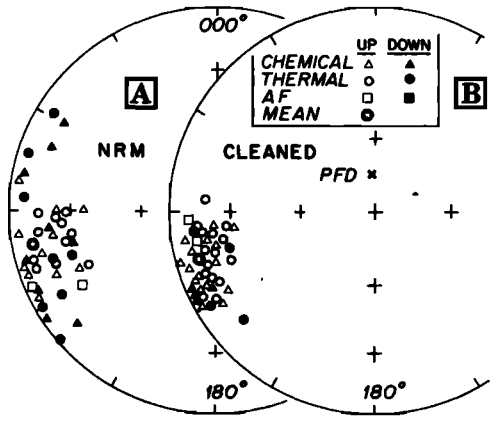


Fig. 2. Direction of magnetization (*a*) before and (*b*) after cleaning treatment (usually 6700 hours, 675°C, or 290 mT) of 41 specimens (eight sites) of section SA. Results from site 5 are not included but are shown in Figure 5. The site mean directions are given in Table 1 and shown in Figure 6. PFD is the present field direction. Equal area polar projection.

1969]. The af-treated specimens were demagnetized in seven or eight steps up to 290 mT. All thermally treated specimens were demagnetized in steps of 100°C up to 500°C and thereafter in steps of 50°C and 25°C up to 675°C; some were heated up to 685°C. The chemical leaching consisted of immersing the specimen in hydrochloric acid (HCl) for a given time, rinsing it in water, storing it in quasi-zero field for 4 days, measuring it, and reimmersing it in fresh acid. The acid strength was 8 *N* for the first 100 hours and 10 *N* thereafter. The number of steps taken to reach an accumulated time of 6700 hours varied from 10 to 14 for different groups of specimens. The accumulated times (steps) are indicated in the figures. In some graphical representations (Figures 5, 8, 10, and 11) the results obtained after some of the cleaning treatments have been omitted in order to facilitate the reading of the representations. In those instances the omitted directions would lie on the path joining the directions shown. The first direction shown is always the natural remanent magnetization (NRM) direction. The measurements were performed by using an astatic magnetometer with negative feedback which can readily detect nonuniformity of direction of magnetization within a specimen [*Roy et al.*, 1972].

During the course of the investigation, numerous specimens were sliced into halves at different stages of cleaning for one of two reasons: (1) to observe acid penetration within the specimen or (2) because two directions of magnetization were clearly present within that specimen. Unless it is otherwise indicated in the figures, the following symbols have been adopted to identify the treatments: circles, thermal; triangles, chemical; and squares, af. Open symbols indicate negative inclinations, and closed symbols positive inclinations. The number identifying the direction changes and remanence decay curve in the figures refers to the site to which the specimen belongs. This facilitates comparison of the remanence behavior during cleaning with the statistics obtained for that site as given in the tables.

RESULTS

The results are similar to those obtained by *Roy and Park* [1974], who proposed that the remanence acquisition of red beds is a lengthy process which begins with the deposition and alignment of magnetic particles in the ambient field (early phase), is followed by hematite growth by authigenesis (intermediate phase), and terminates with the compaction and cementation during lithification (late phase). The three phases

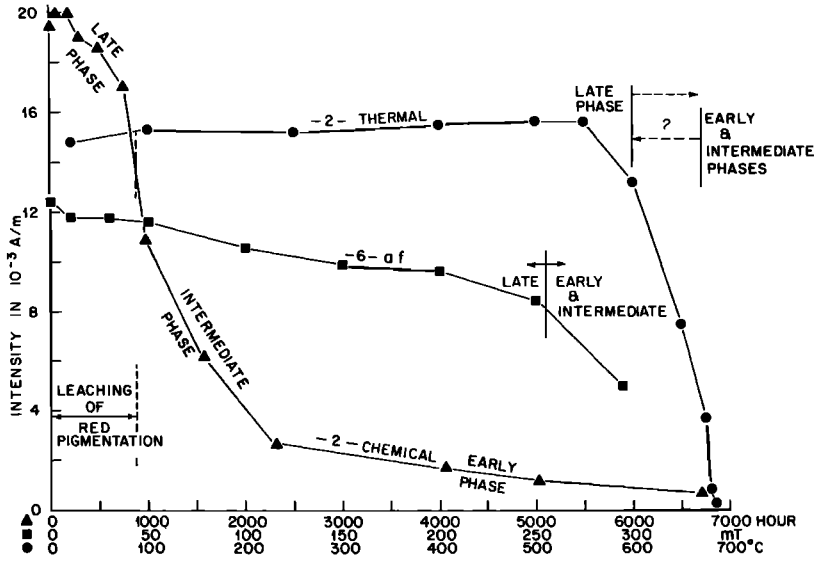


Fig. 3. Intensity decay curves obtained during chemical, thermal, and af treatments. The three-phase model shown is based on and can explain all the results from three red bed studies, this study and those of *Roy and Park* [1974] and *Roy* [1977]. In all three studies the transition between the late and the intermediate phase always coincides with the disappearance during leaching of the pigment, usually red but sometimes a different color (see Table 2). The transition between the intermediate and the early phase usually takes place at the change of gradient of the decay curve. Thermal treatment appears to remove the late phase remanence first; this usually occurs between 600°C and 670°C and varies amongst specimens. Treatments at higher temperatures (in steps of 5°) appear unable to separate the intermediate and late phase remanences. All remanence simply disappears between 675°C and 690°C. The af results are less numerous (none from *Roy and Park* [1974]). However, it seems that most of the late phase remanence is removed by fields of about 250 mT, while fields of 290 mT do not affect the intermediate and early phase remanences. Examples of the relationship between intensity decay curves and direction changes are shown later (Figures 10 and 11).

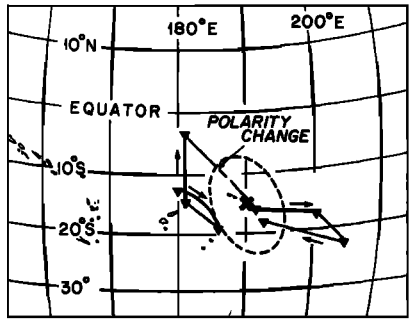


Fig. 4. Site poles of the SA section. The result of site 5 has not been used because of the polarity change that took place at that time as indicated. Mean and oval of confidence at $P = 0.05$ are shown.

could be recognized and sequentially separated by chemical treatment. Because of polarity changes between phases, it could be shown that the early and intermediate phases had similar blocking temperature (T_B) spectra confined to a short interval near the Curie point of hematite and consequently could not be separated thermally. It was also established that the late phase remanence was carried by the red cement and had T_B distributed over a wide range from 20°C to 600°C and sometimes up to 674°C. In another study of red beds [*Roy*, 1977], it was found that the results obtained by performing chemical and thermal experiments were quite consistent with the three-phase model; in addition, af experiments appeared to indicate that most of the late phase remanence can be demagnetized by cleaning in fields of 200–250 mT while the intermediate and early phase remanences remain unaffected by fields as high as 290 mT.

The NRM of the Jacobsville is extremely diverse amongst sites. Sometimes the within-site directions are fairly tightly grouped, but three mean directions are found: a westerly direction and an easterly direction, both with shallow inclinations, and a northerly direction, with a steep positive inclination. At other sites the directions are so scattered that they appear meaningless. Detailed experimental work indicates that this is owing to the superimposition of remanence phases. At all but one site, three phases of magnetization can be recognized by chemical leaching but not by thermal and/or af treatments, in accordance with previous findings. The three-phase model is therefore used throughout this study to explain the results and to determine the relative effectiveness of the three cleaning techniques.

THE SA SECTION

Nine sites were sampled from a section (SA) near the town of Jacobsville. The beds consist of flat-lying fine-grained red sandstones; 5.3 m of that section could be reached for sampling, and the stratigraphic separation of two adjacent sites is given in Table 1.

At eight of the nine sites it appears that most if not all of the magnetization was acquired in a field of single polarity (Figure 2). Treatment clearly improves the grouping of directions (compare Figures 2*a* and 2*b*). The precision of the between-site mean also increases considerably: from 14 to 42 (Table 1).

The applicability of the three-phase model is clearly demonstrated by the chemical experiments (Figure 3). The transition between the late and the intermediate phase coincides with the disappearance of the red coloring which occurs between 495 and 1000 hours, as was verified by slicing a number of specimens after the 495-, 740-, and 1000-hour treatments. The transition between the intermediate and the early phase occurs at the leveling of the remanence decay curve after about 2500 hours of leaching. As was mentioned before, all three phases cannot be recognized by thermal or af treatment. However, there are several indications that the magnetization removed by thermal treatment at 600°C–670°C and af treatment at 250 mT is that of the late phase carried by the red pigment. Most of the direction changes (Figure 2) taking place during treat-

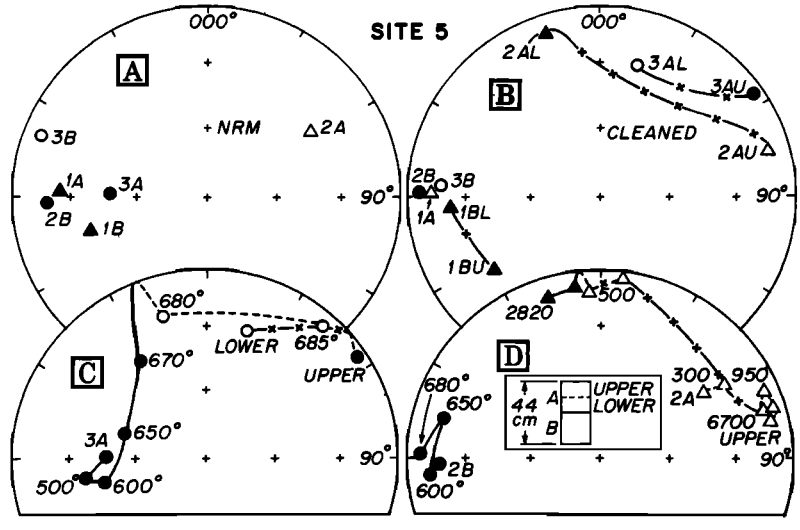


Fig. 5. Results at site 5 where a reversal occurred. Directions (*a*) before and (*b–d*) after are shown. Specimens A and B indicate specimens from the same core, A being above B. L and U indicate the lower and upper half of a specimen that has been sliced during the treatment; they are joined together or to the whole specimen by lines broken by crosses. The last treatment was 680° for 2B, 685° for 3A and 3B, and 6700 hours for all chemically treated specimens except 2A lower (2820 hours; see Figure 5*d*); the remanence after 5100 hours of leaching was too weak to be measured. Specimen 2 (*d*) clearly shows the reversal, 2A upper and 2B being oppositely directed. Slicing of 3A at 685° shows that the directions of the two halves are quite different. None of the halves, however, yields the field direction before or after field reversal. This is attributed to unresolved intermediate and early phase remanences. The mean direction of site 5 after treatment given in Table 1 has been obtained by excluding 2AL and 3A and averaging 1BL and 1BU. For clarity of representation, directions obtained after some of the intermediate cleaning steps are not shown; see the Figure 8 legend for details.

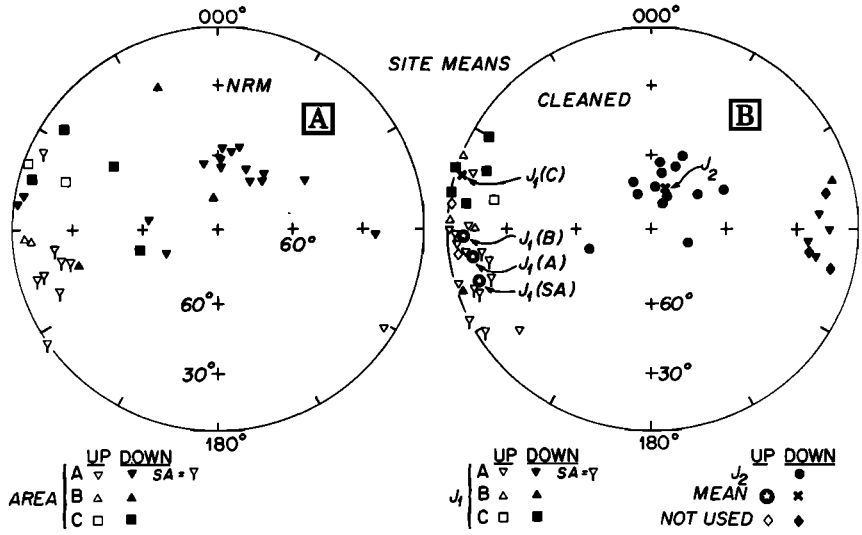


Fig. 6. All site mean directions (except site 5) before treatment are shown in Figure 6*a*; the areas from which they originate are shown by different symbols. In Figure 6*b* the J_1 and the J_2 site mean directions are shown. The areas of J_1 results are identified. The mean direction for each area is shown; for calculation of these area mean directions the site mean directions (diamonds) obtained from two or fewer observations ($N_1 \leq 2$; Table 2) have been excluded. Site identification of J_2 site mean direction can be obtained from Table 2.

ment usually occur in the first 1000 hours of leaching (during removal of the red pigment) and correspondingly below 600°C–670°C and below 250 mT. Also, the red pigment accounts for about 25% of the remanence (Figure 3); the thermal and af decay curves show that about the same percentage of remanence has blocking temperatures T_B ranging from 20°C to 600°C–650°C and remanent coercive forces (rcf) below 250 mT. These observations are in accordance with results from other red bed studies [*Roy and Park*, 1974; *Roy*, 1977]. The suggested T_B and rcf ranges of the late phase are shown in Figure 3. Evidently, in this instance it is difficult to determine exactly the upper limit of those ranges, which also probably varies amongst specimens. The uncertainty in the determination of those limits is shown in Figure 3. In an attempt to determine the direction of magnetization of the late phase, vector subtraction was performed on all 41 specimens. This gives a direction of 274°, +03° (k = 5; α_{95} = 11°) yielding a pole at 179°E, 04°N. This result should be taken with reservation and regarded as a very rough estimate. The differences between the two values used for vector subtraction were small in both direction and intensity; the mean direction (of this late phase remanence) is only 24° away from the mean direction obtained after cleaning (Table 1), and the intensity differences accounted for only about 25% of the total remanence. The accuracy of a vector analysis performed under these circumstances is questionable, as is shown by the small k.

The site mean poles obtained after treatment (Figure 4) show that considerable field direction changes took place during magnetization of the sampled section. An F distribution test [*Watson*, 1956] gives F = 5.9 and indicates that the between-site dispersion is significant. This means that the sites were magnetized sequentially and that the path of Figure 4 represents the pole movement relative to the sampling locality. It is noted that this path greatly resembles the path of secular variation of the recent field [*Thellier*, 1966]. The fact that the path forms a closed loop indicates that the variation has been effectively averaged out. The pole (190°E, 16°S) obtained from the mean (Table 1) is then regarded as a good estimate of the average pole location at the time of or shortly after deposition of this section.

A change of polarity of the field evidently occurred during that time, as evidenced at site 5 (Figure 5). This polarity change probably was (1) of short duration, since it was recorded by only two specimens, and (2) rapid, since two superimposed specimens (2A and 2B) from a single core are oppositely magnetized. Slicing of 2A emphasizes the rapidity of the reversal. The direction of 2A upper is clearly to the east, and that of 2B to the west. The direction of 2A lower may be the direction of the field during transition or the direction of the resultant vector of an easterly and a westerly directed magnetization.

Such rapid reversals of short duration have been recorded in other studies [*Roy and Park*, 1974; *Larson and Walker*, 1975; *Roy*, 1977], and it is becoming evident that numerous reversals could easily remain undetected. This creates serious problems for the erection of polarity scales and/or correlation of different rock units on the basis of polarity of magnetization, as has been suggested for the Keweenawan rocks [*Palmer*, 1970; *Robertson and Fahrig*, 1971]. Even in the ideal case of a stratigraphic succession of continuously deposited sediments, closely spaced sampling may be required to detect all polarity changes that have occurred during the time of deposition. In this instance, one reversal was detected because site 5 happened to be sampled although it is only 0.9 m below and above sites 4 and 6, respectively. Many other reversals may be recorded in that 5.3-m section.

Other Sites

Specimens (210) from each of the remaining 28 sites were subjected to one of the three treatments: chemical (80 specimens), thermal (86 specimens), or af (44 specimens). The site mean results are shown in Figure 6 and Table 2. The areas from which the sites originate are identified by symbols; the site mean directions of the SA section are also shown for comparison and completeness. The NRM data show results prior to treatment of any kind. The cleaned site means show the presence of two very stable remanences: the J_2 remanence (shown by solid circles irrespective of the area; see Table 2 for identification) is steeply inclined and oriented toward the north (021°, +74°), and the J_1 remanence, with a shallow

TABLE 2. Site Mean Results Before (NRM) and After Treatment (J_1 and

Site	NRM				J_2 Remanence				
	Number of Specimens N	$D°, I°$	Precision Estimate k	Radius of Circle of Confidence α_{95}	Number of Determinations N_2	$D°, I°$	Precision Estimate k	Radius of Circle of Confidence α_{95}	Pole
Area A									
10	13	349, +63	27	8	13	339, +70	19	10	348°E, 65°N
11	11	035, +67	10	15	9	110, +74	7	21	305°E, 31°N
12	8	092, +24	11	17					
13	14	014, +56	40	6	14	023, +58	12	12	017°E, 71°N
14	7	003, +57	11	20	6	011, +67	11	22	337°E, 82°N
15	6	010, +57	9	23	6	025, +76	24	14	301°E, 76°N
16	6	246, +67	2	59					
17	8	061, +50	9	20	7	062, +56	4	32	352°E, 44°N
18	8	280, +05	15	15					
19	7	277, +02	47	10					
20	6	277, +62	2	70					
21	6	121, −06	1	95					
22	12	041, +64	10	14	10	026, +78	9	7	298°E, 66°N
23	5	003, +65	106	7	5	340, +75	22	17	241°E, 71°N
24	6	025, +64	35	12	5	022, +63	18	19	001°E, 74°N
25	4	041, +63	18	22	4	054, +66	10	30	339°E, 54°N
26	11	001, +62	45	7	11	007, +63	18	11	028°E, 84°N
Area B									
27	7	338, +26	6	26					
28	7	268, −07	4	33					
29	11	352, +77	18	11	10	006, +73	21	11	286°E, 77°N
30	6	256, +27	13	19					
31	6	267, −08	11	24					
Area C									
32	5	288, −24	6	35					
33	3	301, +40	17	31					
34	4	303, +11	25	19					
35	11	255, +57	22	33	10	251, +64	7	19	231°E, 22°N
36	5	285, +06	50	11					
37	7	290, −02	10	20					

*The e and w indicate the number of easterly and westerly J_1 determinations.
†Sites with $N_1 \leq 2$ have not been used to calculate the area A mean shown in Figure 6 and Table 3.

inclination, has dual polarities, the westerly directed (J_1w) and the easterly directed (J_1e).

The J_1 remanence was found at 27 sites, and the J_2 remanence at 13 sites; the two remanences were found to coexist at 12 sites. A clear separation of the two remanences and an accurate determination of each could not be realized in all specimens owing to (1) the inefficiency of the cleaning technique used, (2) the dual polarities of J_1, (3) the intensity ratio of J_2/J_1 being near or less than unity, and (4) the overlapping of the T_B or rcf spectra of J_2 and J_1. Factors 1, 2, and/or 4 often prevented the determination of J_1, and 1, 3, and/or 4 sometimes made the determination of J_2 unreliable. The numbers of J_1 and J_2 determinations are given under N_1 and N_2, respectively, in Table 2. The J_1 data represent results obtained after extensive cleaning, generally 6700 hours, 675°C–685°C, or 290 mT. The J_2 data were obtained by vector subtraction of the last cleaning step (generally 6700 hours, 675°C–685°C, or 290 mT) results (V_1) from the 100-hour, 600°C, or 200-mT results (V_2). The methods used for the separation of J_1 (both polarities) and J_2 are now described; it is simpler to describe the separation of J_2 first.

THE J_2 REMANENCE

At 13 sites, J_2 is clearly predominant and is carried by most if not all specimens. Chemical leaching shows that remanences J_1 and J_2 coexist at 12 sites and that the J_2 carriers are less resistant to leaching than the J_1 carriers; this is evidenced by a complete shift from the J_2 direction to the J_1 direction. Thermal and af treatments, although they are less effective for the separation of the two remanences, also show that they are both present and have high T_B and rcf. Upon cleaning, the direction

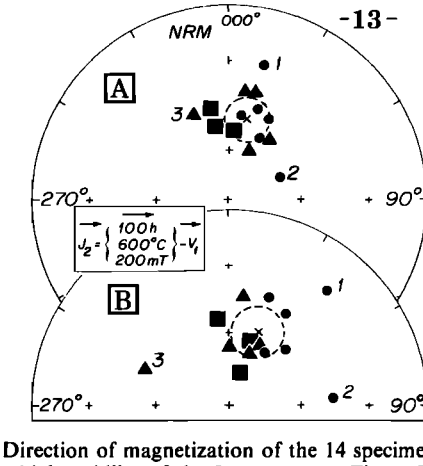


Fig. 7. Direction of magnetization of the 14 specimens of site 13 showing the high stability of the J_2 remanence. Figure 7*a* shows the direction prior to the cleaning treatment. Figure 7*b* shows the direction of the magnetization remaining after the 100-hour, 600°C, or 200-mT treatment minus (vectorially) the magnetization remaining after the last cleaning treatment, usually 6700 hours, 675°C, or 290 mT.

J_2 Remanences) of All Sites Except the First Nine Sites Listed in Table 1

J_1 Remanence						Samples	
Number of Determinations N_1	e, w*	$D°, I°$	Precision Estimate k	Radius of Circle of Confidence α_{95}	Pole	Grain Size	Color
9	8, 1	243, −21	8	19	195°E, 26°S	fine–medium	red
5	0, 5	099, +20	11	24	178°E, 01°S	coarse	brown
8	0, 8	092, +24	11	17	171°E, 08°S	very coarse	red
7	3, 4	255, −02	14	17	193°E, 11°S	very coarse	brown
2†	1, 1	278, −02				coarse	red
1†	0, 1	099, +21				medium	red
3	3, 0	270, −01	28	22	180°E, 00°	medium	red
3	0, 3	085, +19	23	26	179°E, 11°S	medium	red
7	7, 0	266, −06	81	7	182°E, 05°S	fine	red
7	7, 0	271, −12	41	9	176°E, 03°S	fine	red
4	4, 0	287, −10	18	22	165°E, 08°S	coarse	brown
4	1, 3	090, +14	14	26	176°E, 06°S	coarse	brown
4	4, 0	263, −10	34	16	184°E, 08°S	coarse	brown
1†	1, 0	264, −05				medium	beige
1†	0, 1	077, +14				medium	beige
1†	0, 1	102, +11				medium	beige
						medium	brown
6	6, 0	291, −03	12	19	166°E, 13°N	medium	pink
6	6, 0	272, 00	13	19	181°E, 01°N	medium	pink
3	1, 2	075, +10	18	30	190°E, 14°S	medium	red
6	6, 0	252, +02	17	16	197°E, 11°S	fine	pink
6	6, 0	270, −15	14	18	178°E, 05°S	fine	pink
5	5, 0	280, −23	16	20	170°E, 02°S	fine	beige
3	3, 0	289, +16	23	26	177°E, 19°N	fine	grey
4	4, 0	300, +10	46	14	166°E, 24°N	fine	grey
7	6, 1	278, +10	9	21	184°E, 03°N	medium	pink
5	5, 0	287, +01	53	11	173°E, 12°N	fine	pink
6	4, 2	280, +00	9	23	178°E, 07°N	fine	pink

moves away from the J_2 direction and toward the J_1 direction without necessarily reaching it. This indicates that although the T_B and rcf spectra of J_1 and J_2 may possibly overlap, the spectra of J_2 are lower than the spectra of J_1, meaning that vector analysis may be of some use for the definition of J_2.

To ensure that J_2 was stable (high T_B and high rcf), the 100-hour, 600°C, or 200-mT results ($\mathbf{V}_2$) were used as 'basic' data. To reduce the effect of the J_1 remanence, the results from the last cleaning step were subtracted vectorially ($\mathbf{V}_2 - \mathbf{V}_1$). For the chemically treated specimens, where J_2 and J_1 can be separated, this provides one with a good estimate of J_2. For the thermally treated and af-treated specimens, where J_2 and J_1 cannot be separated, $\mathbf{V}_2 - \mathbf{V}_1$ does not yield a precise determination of J_2. However, in most instances, $\mathbf{V}_1$ is directed toward the west or east with a low inclination (<35°), indicating that it is composed of J_1 predominantly; therefore $\mathbf{V}_2 - \mathbf{V}_1$ is a closer approximation of J_2 than $\mathbf{V}_2$ would be. Vector analysis was performed on all specimens from these 13 sites. For most specimens the difference between $\mathbf{V}_2$ and $\mathbf{V}_1$ was large, and the number of determinations per site is given in Table 2 under N_2. Only in very few specimens (N-N_2 < 2; Table 2) was the difference between $\mathbf{V}_2$ and $\mathbf{V}_1$ quite small, making determination of J_2 (if present) very imprecise; these of course were not used for calculation of the site mean direction.

At the other 15 sites, J_1 was predominant, and the difference between $\mathbf{V}_2$ and $\mathbf{V}_1$ was not sufficient to establish whether J_2 was present. However, at many sites there is a marked increase in k upon cleaning; comparison of NRM with J_1 shows a twofold to fivefold increase at sites 18, 28, and 32 and drastic increases at sites 20 and 21. These increases might be attributable at least in part to the removal of a J_2 remanence.

It should be noted that for all sites the level used for separating J_2 and J_1 was standardized to 100 hours, 600°C, and 200 mT. A higher within-site k could have been obtained by using selective levels of separation for each specimen. This is demonstrated at site 13 (Figure 7), where the directions of three specimens (1, 2, and 3) have already started to move toward J_1e or J_1w. The use of the 550°C results for 1 and 2 and the 24-hour results for 3 increases k from 12 to 19. Standardization greatly simplifies computation and presentation of the results, and although it produces poorer looking statistics, it probably does not affect greatly the overall site-mean-averaged direction (021°, +74°).

The J_2 remanence has T_B and rcf of at least 600°C and 250 mT and is never removed prior to the disappearance of all staining which occurs between 100 and 750 hours, depending on the porosity of the rock. These are characteristics of late phase remanence carried by the pigment on the three-phase model (Figure 3). Of much interest is the fact that in many specimens, J_2 is not removed until 1520–2300 hours of leaching, long after removal of the staining. It then appears that in hose specimens the acquisition of J_2 was not limited to the late

phase but began during the intermediate phase. This of course implies that J_2 was part of the initial remanence acquisition process. This interpretation will be discussed in more detail, taking into consideration all possible alternatives, after presentation of the J_1 results.

THE J_1 REMANENCE

The J_1 remanence (J_1w and/or J_1e) can be recovered from all sites of the three areas sampled except site 26 (Table 2). It is evident that at least one reversal took place during the magnetization process of J_1. t certain sites (19, 30, 31, 32, 34, and 36), all specimens carry a J_1w remanence, indicating that the field was westerly directed during most of the process; at site 12, all specimens carry a J_1e remanence, indicating an easterly directed field. At other sites (10, 13, 14, 21, 29, 35, and 37), both J_1w and J_1e are found.

The occurrence of reversal(s) accounts for the fact that field direction determinations could not be obtained from all specimens (Table 2; $N_1 < N$). Because the magnetization process can last a few million years and consequently can span field reversal time scales [*Roy and Robertson*, 1968; *Roy and Park*, 1974; *Larson and Walker*, 1975], the magnetization within a specimen can be acquired partly before (M_b) and partly after (M_a) a field reversal. In those instances the direction of the observed resultant vector ($\mathbf{M}_b + \mathbf{M}_a$) will be different from that of the field unless M_b and M_a are exactly antiparallel, which is rare if only because of secular variation. When the intensities of M_b and M_a are comparable, the angular difference between the direction of the observed vector and that of the field may be very large (up to perpendicularity). To obtain a field direction determination, it is therefore necessary to separate M_b and M_a. Different techniques and analytical procedures previously used in a number of red bed studies [*Roy and Park*, 1974; *Roy and Lapointe*, 1976; *Lapointe*, 1978; *Morris*, 1977*a*, *b*; *Roy*, 1977] have been applied to the Jacobsville, and the results again demonstrate the limitations of each technique for the unraveling of the different phases of magnetization.

The thermal and af cleaning techniques are ineffective for the separation of M_b and M_a. After the initial displacement produced by removal of J_2 the specimen mean direction often comes to rest some distance away (referred to as the intermediate direction) from the site mean direction, indicating that the T_B (680°C; Figure 8*b*) and rcf (250 mT; Figure 8*c*) spectra of M_b and M_a are similar and cannot be separated.

Chemical leaching is clearly more successful than thermal or af demagnetization. When two specimens from the same core are treated, one thermally (or by af) and the other chemically, the specimen direction of the latter often reaches the site mean direction (Figure 8*a*), while the specimen direction of the former does not and remains intermediate (Figure 8*b*). The opposite never occurs. Therefore even though M_b and M_a have similar magnetic characteristics (T_B and rcf), they can be separated chemically. Gradually longer leaching times will eventually destroy the magnetic grains carrying one of the remanences without attacking the grains carrying the other, so that the latter can be isolated. The fact that the two remanences behave differently under chemical leaching and have different directions indicates that they were produced under different conditions during the remanence acquisition process and belong to different phases, that is, the early and intermediate phases (the late phase having been associated with subsequent pigmentation). According to the more detailed work of *Roy and Park* [1974] the older M_b is the more resistant to leaching and would be produced during the early phase, which takes place during and/or shortly after deposition.

By chemically isolating the early phase remanence, field direction determinations were often (Figure 8*a*), but not always (Figures 8*d* and 8*e*), obtained. The chemically treated specimen will yield the field direction only if the reversal was post-early-phase. When the reversal occurs during the early phase, the upper and lower parts of the specimen will have different remanence directions, and the specimen mean direction will be more or less different from the field direction according to the intensity ratio of M_b/M_a. This is easily verified by slicing the specimen into halves along the bedding. The examples (Figures 8*d* and 8*e*) show that the preslicing intermediate direction (500 hours) is the direction of the resultant vector of two magnetizations whose directions have a very large angular difference. Usually, the magnetization direction of one half will be near the site mean direction, while the magnetization direction of the other half will tend toward but will not reach the site mean reverse direction. Both directions will rarely reach the pre- and post-field-reversal mean direction, respectively; this requires the slicing to coincide exactly with the polarity transition.

The presence of two differently directed magnetizations within a specimen is readily observed with a direct read astatic magnetometer. The declination of the magnetic vector is measured by placing the specimen in an upright ($D_1°$) and an inverted ($D_2°$) position under the lower magnet. Inclination measurements ($I_1°$ and $I_2°$) are obtained with the specimen in two positions about its axis and positioned so that the mean horizontal component rotates in a horizontal plane. A qualitative measure of within-specimen uniformity of direction ($\Delta\Psi°$ [*Roy and Fahrig*, 1973]) is obtained by using the empirical formula

$$\Delta\Psi° = \left[\frac{(\Delta D° \cos I_m°)^2 + \Delta I°^2}{2}\right]^{1/2}$$

where $\Delta D° = D_1° - D_2°$, $\Delta I = I_1° - I_2°$, and $I_m° = (I_1° + I_2°)/2$. For example, the $\Delta\Psi$ values for the specimen from site 21 (Figure 8*d*) were as follows: before slicing, 89°; after slicing, westerly directed, 16° at 500 hours and 14° at 6700 hours; and after slicing, easterly directed, 45° at 500 hours and 22° at 6700 hours. This method is by far the most effective for obtaining field direction determinations. It has also been applied to thermally treated specimens (Figures 8*f* and 8*g*); the results are not as satisfactory. Because the early and intermediate phase remanences have not been separated, the difference between the magnetization direction of either half and the site mean direction is usually larger than when chemically treated specimens are used, and of course it is impossible to determine if the field reversal occurred during the early or intermediate phase.

When $\Delta\Psi° < 30°$ was arbitrarily used as a means of filtering out unresolved $\mathbf{M}_b + \mathbf{M}_a$, 124 field direction determinations were obtained from over half of the 210 specimens (Table 2). Most of those determinations were obtained from chemically treated and/or sliced specimens, some from thermally treated specimens, and very few from af-treated specimens. Undoubtedly, more determinations would have been obtained if all specimens had been treated chemically. However, the use and comparison of the three cleaning techniques provide one with more information about the different phases of magnetization, such as T_B and rcf.

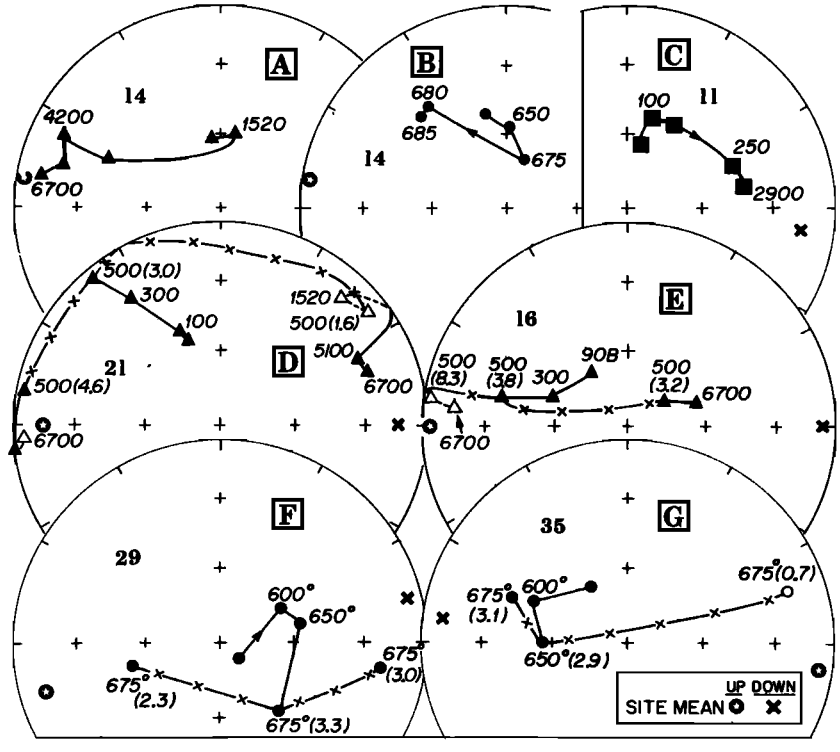


Fig. 8. Direction changes during treatment. The mean direction of the site from which the specimen originates is shown. With (*a*) the chemical treatment the direction often reaches the site mean direction. With (*b*) thermal or (*c*) af it rarely does. (*d–g*) The specimens have been sliced during the treatment; the halves are joined to the whole specimen by lines broken by crosses. The numbers in the brackets following treatment identification are the intensities in 10^{-3} A/m before and after slicing. The vector sum of the two halves, although it does not differ by much, is not equal to the vector of the whole specimen for two reasons: (1) the slicing causes the loss of a 3-mm thickness of the specimen, and (2) the preslicing measurement is the result of integration of all parts of the specimen as opposed to the measurements of two separate halves. Because the early phase remanence can be isolated by chemical leaching, the direction of at least one half specimen will eventually reach the site mean direction (*d* and *e*). This rarely occurs for the thermally treated specimens because the early and intermediate phase remanences cannot be separated by means of this treatment. To facilitate the reading of the graphical representations, some of the intermediate directions have been omitted. Their plotting would have overloaded the stereonets unduly. For example, in Figures 8*d* and 8*e* there were eight cleaning steps between 500 and 6700 hours; the eight directions lie on the line drawn from 500 to 6700 hours.

AGE OF J_1 REMANENCE

The mean direction, pole position, and statistics for each of the areas sampled are listed in Table 3, and the poles plotted in Figure 9 with other relevant poles. It is suggested that the southward track drawn in Figure 9 represents the pole movement relative to North America between approximately 1100 and 1040 m.y. ago. The upper portion belongs to the lower end of the south trending Keweenaw Track (shown in more detail in Figure 12). The Cardenas Lavas (pole CL [*Elston and Grommé*, 1974]) have a Rb-Sr isochron age of 1090 ± 70 m.y. [*McKee and Noble*, 1976]. The Nonesuch Shale (NO) and Freda Sandstone (FR) poles [*Henry et al.*, 1977] derived from thermal and chemical work confirm the trending toward the south and down to the equator; these poles have been assigned ages of 1060 and 1050, respectively, by the authors. The well-defined J_1(SA) pole (Table 1) extends the track further to the south (down to 16°S) and indicates that the Jacobsville on the Keweenaw Peninsula is younger than the Freda Sandstone of Michigan and would appear to be 1050–1000 m.y. old.

TABLE 3. Composite Site Mean Results Calculated From Tables 1 and 2

Remanence	Area	Number of Determinations N	$D°, I°$	Resultant Vector R	Precision Estimate k	Radius of Circle of Confidence α_{95}	Pole	$\delta p, \delta m$*
J_1	SA	8	254, −12	7.8	42	9	190°E, 16°S	5, 9
	A	19	262, −12	18.42	31	6	184°E, 10°S	3, 6
	A†	24	264, −12	23.26	31	5	182°E, 09°S	3, 5
	B	5	268, −05	4.82	23	16	182°E, 03°S	8, 16
	A + B	24	263, −11	23.19	29	6	183°E, 09°S	3, 6
	C	6	286, +02	5.81	26	13	175°E, 12°N	6, 13
J_2	A + B + C‡	13	021, +74	12.47	23	9	307°E, 72°N	13, 16

*The δp, δm are the semiaxes of the oval of confidence (P = 0.05).
†Including sites with $N_1 \leq 2$ in Table 2.
‡Site locations corrected to 271.6°E, 47°N.

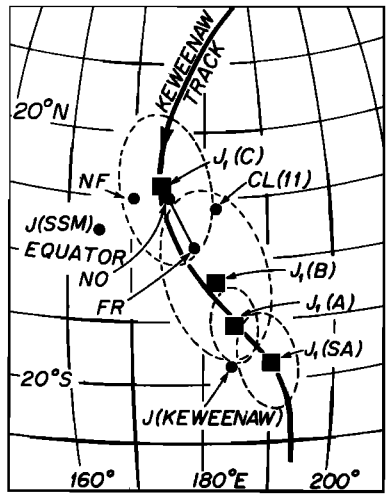


Fig. 9. The apparent polar path for North America. It is suggested that the time interval represented is −1100 to −1040 m.y. approximately. The J_1 poles from the C, B, and A sampled areas and the SA section are shown with approximate ovals of confidence (P = 0.05). The Nonesuch-Freda (NF), Sault Ste. Marie Jacobsville (J(SSM)) and Keweenaw Peninsula Jacobsville (J(KEWEENAW)) poles are from *Dubois* [1962]; the Nonesuch (NO) and Freda (FR) are from *Henry et al.* [1977]; and the Cardenas Lavas (CL(11)) are from *Elston and Grommé* [1974].

J_1(C) appears to be considerably older than J_1(SA). An F distribution test for comparison of mean directions [*Watson*, 1956] indicates that J_1(C) and J_1(SA) do not belong to the same population and are significantly different (Table 4). The test shows that J_1(C) also differs from J_1(A) and J_1(B). Only J_1(B) and J_1(A) are not statistically different, but this probably reflects the poor isolation of J_1(B) as indicated by the oval of confidence. Because of this uncertainty, J_1(A + B) is also given in Table 3. However, it should be noted that isopach maps [*Hamblin*, 1958] support a J_1(C) → J_1(B) → J_1(A) → J_1(SA) sequence. This is also in accordance with the fact that the southernmost pole J_1(SA) has been obtained from the uppermost Jacobsville [*Babcock*, 1975].

The clustering of the J_1(C), NO, FR, and CL poles indicates that these rock units may be contemporaneous. A possible correlation between the Sault Ste. Marie Jacobsville and the Michigan Nonesuch-Freda was first advanced by *Dubois* [1962] on the basis of the agreement between the J(SSM) and NF poles. Compressional wave velocity measurements also indicate a great similarity between the Sault Ste. Marie Jacobsville and the Wisconsin Freda rocks, while the Michigan Jacobsville rocks have considerably lower velocities [*Halls*, 1969].

It is suggested that the Sault Ste. Marie sandstones (Jacobsville?) are older than the Keweenaw Peninsula Jacobsville, approximately equivalent to the Nonesuch-Freda of Michigan, and perhaps as old as 1090 m.y. based on the age of the Cardenas Lavas pole.

Time of Acquisition of J_2

The site mean direction of J_2 is near the direction of the present earth's field (Table 2; Figure 6). Therefore it might be very tempting to explain J_2 simply as a remagnetization in the present field. However, since J_2 is a very hard magnetization which resists 100 hours of leaching, thermal treatment at 600°C, and af treatment in 200-mT fields, it seems that three possible interpretations should be considered. J_2 represents (1) the resultant vector of two unresolved magnetizations, J_1e and J_1w, (2) part of the initial process of magnetization, or (3) a secondary magnetization acquired in the present field.

Interpretation 1 could possibly be applicable to a few sites with low k (11, 17, and 35). However, at most sites the within-site grouping is fairly tight (Figure 7), and it is difficult to imagine how two oppositely directed magnetizations could be combined to produce a resultant vector direction so consistently amongst so many specimens. Most importantly, at that level of cleaning (100 hours, 600°C, 200 mT), the $\Delta\Psi°$ is still usually small (<20°), an indication that the predominant magnetization is uniformly directed throughout the specimen. A plausible explanation for the low k at sites 11, 17, and 35 is that a good portion of J_2 has already been removed. This is in accordance with the fact that low precisions (k) are found at sites where the specimens are coarse grained.

Interpretation 2 is logical if a relationship can be established between the specimens that carry J_2 and those that do not. One test is to compare the remanence decay curves of specimens (group B) carrying the steeply inclined J_2 with those of specimens (group A) carrying only J_1 (all phases of magnetization occurred in a westerly and/or easterly directed field). If J_2 was part of the initial process, then the magnetic characteristics of group A and group B specimens should be comparable.

Remanence decay curves from all three cleaning techniques are shown in Figures 10 and 11; group B specimens can be identified on the stereonets by steep inclinations prior to cleaning. The similarity of the decay curves is remarkable. In fact, in the thermal and af curves (Figure 11) the two groups would be indistinguishable if normalized. The chemical decay curves (Figure 10) clearly show that three phases of magnetization are present in all specimens. The leaching time required for separating the intermediate and early phase remanences and isolating the latter is about the same (≃2800 hours) for both group A and group B specimens. The leaching time for the removal of the late phase remanence is usually shorter for group B (150–740 hours) than for group A (495–1000 hours) because group B specimens are in general coarser grained. Nevertheless, the transition between the late and the intermediate phase always coincides with the disappearance of the pigmentation. Of particular interest for this interpretation (and definition of the polar path later) is the timing of the direction changes from J_2 to J_1. In some instances (13 in Figure 10*a*), J_2 is removed after 150–300 hours simultaneously, with discoloration indicating that it is carried by the pigment exclusively and that the change into the J_2 direction occurred prior to the late phase. In other instances (24 and 33 in Figure 10*b*), direction changes do not occur until 1520–2300 hours of leaching, indicating that the shift took place prior to or during the intermediate phase. The variations (amongst specimens) of the timing of direction changes indicate that all phases were part of the initial process of remanence acquisition and that a large apparent polar movement took place during the process. A gradual polar movement is also seen at site 13 (Figure 10*a*). The J_1 direction is not reached until 5100 hours; from 300 to 2900 hours the direction is clearly intermediate between the J_1 and J_2 directions. A shift to the J_2 direction during the intermediate phase also explains the very high T_B (up to 675°; Figure 11*a*) and high rcf (290 mT; 23 in Figure 11*b*) of J_2 found in many specimens.

Interpretation 3 is based solely on the similarity between the direction of J_2 and the direction of the present field, and it is assumed that J_2 was acquired recently. This interpretation meets with many difficulties. Firstly, there is the selectivity;

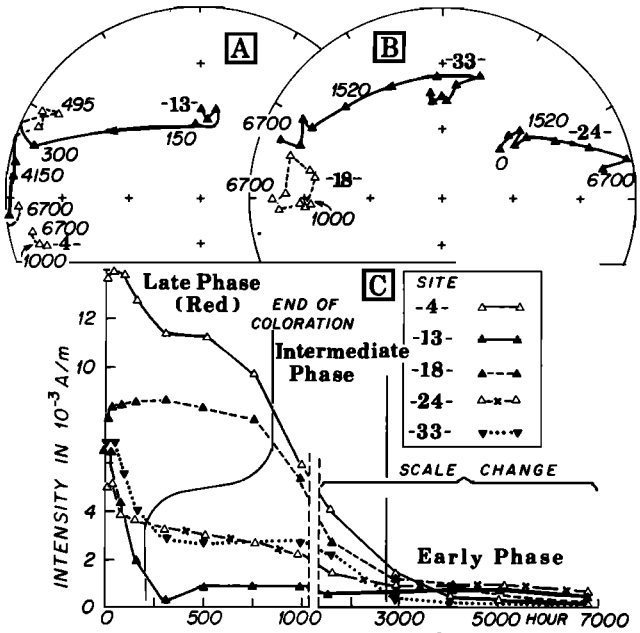


Fig. 10. Comparison of remanence decay curves of chemically treated specimens carrying the steeply inclined J_2 (group B) with those of specimens carrying only J_1 (group A). Group B specimens are identified by the steep inclinations (>40°) prior to cleaning in Figures 10*a* and 10*b*; representation of some intermediate directions has been omitted for reasons given in the Figure 8 legend. The cleaning steps are shown in Figure 10*c*. Because of the similarity of the decay curves it is suggested that J_2 was part of the initial process of remanence acquisition.

only 13 of 37 sites would have been remagnetized. Secondly, it must be argued that at those 13 sites the late phase never occurred or it was removed by some unknown mechanism and replaced 1100 m.y. later by another magnetization with characteristics similar to those of the late phase remanence. This is a strenuous argument which still does not explain the J_2 intermediate phase remanence. Thirdly, a few (eight) steeply and negatively inclined NRM's were found at some sites, indicating that a polarity reversal might have occurred during acquisition of the J_2 remanence. Fourthly, and most importantly, a recent origin for J_2 is not supported by the lithogenesis of the Jacobsville. From examination of the leached pattern, *Hamblin* [1958, p. 25] concludes that 'all evidence indicates that the red colour of the Jacobsville formation is a primary feature and that subsequent leaching in selected areas produced the white mottling.'

Conversely, this observation of Hamblin supports interpretation 2, which can best explain all the data. It is therefore concluded that the J_2 remanence was acquired toward the end of the initial process of magnetization.

THE −1100- TO −670-M.Y. PALEOMAGNETIC POLES

Explanation of the Jacobsville results requires a major revision of the apparent polar path during the Hadrynian. According to recent polar paths it has been believed that for at least 500 m.y. (≃1100–600 m.y.) the pole relative to North America remained in the Pacific Ocean, wandering by some 30° about the equator and 170°E. The steeply inclined J_2 direction indicates that the pole did not remain in low latitude but reached high latitude at some time during that period. Other results yielding high-latitude poles can be found in the paleomagnetic literature. Unfortunately, those results have been regularly disregarded. It would appear that any magnetization directed along the present field is arbitrarily attributed to recent remagnetization; this produces a bias toward low-latitude poles that may artificially create a wandering about the equator.

All presently available poles which may fall in the period from −1100 to −670 m.y. are shown in Figures 12, 13, and 14. It is difficult to evaluate the significance of most of those poles because of limited experimental work; in many instances, only af cleaning (<50 mT) was performed on igneous or metamorphic rocks, and many sedimentary units were subjected to thermal and/or af treatments only. Investigations based on such limited cleaning often fail to reveal some of the phases of magnetization [*Roy and Lapointe*, 1978]. Failure to detect one or more phases can, of course, create serious problems in interpretation. For example, if the undetected phase is the initial one, a secondary magnetization could easily be mistaken for the initial. It is, of course, possible that the magnetization revealed after limited treatment is the initial, perhaps because it is the only one carried by the rock. However, because the experimental work was not sufficient to ascertain whether another magnetization was present or not, the result remains uncertain and controversial. Therefore we believe that a minimum of laboratory experiments should be performed on all rock units.

For igneous and metamorphic rocks, both extensive af and thermal treatments are necessary; af alone is inadequate to identify the remanence carriers (e.g., magnetite or hematite). Thermal and chemical remanent magnetization are often found to coexist in such rocks, and either of them could be of initial or secondary origin. Thermal cleaning will at least provide one with the T_B of the different carriers, and it might then be possible to establish the origin and sequence of magnetization [*Buchan and Dunlop*, 1976; *Roy and Lapointe*, 1976]. For sedimentary rocks, thermal and af experiments are insufficient for unraveling complex magnetization sequences; chemical leaching must be performed. For example, the $J_1 \rightarrow J_2$ sequence could not have been established without chemical leaching.

The tracing of the path (Figures 12, 13, and 14) is based on a number of key reference poles (KRP; shown by larger symbols) which are obtained from laboratory experiments meeting the above criteria and which appear to be derived from successfully separated magnetizations. This means that a post-early-phase magnetization (e.g., J_2) can be used to define the trend of the path even though it cannot be used for its calibration. Poles representing the mean of several studies (e.g., Portage Lake Lavas, 392) are also regarded as KRP's. The proposed path therefore relies on a few well-determined poles and especially on magnetization sequences rather than on isolated results of uncertain quality. It is much simpler than previously proposed paths that have required complex looping and/or

TABLE 4. *F* Distribution Test for Comparison of Mean Directions [*Watson*, 1956]

Populations	F^*	$P(F)$ Probability 0.10	0.05	Significant
A, B	1.55	2.43	3.21	no
A, C	15.89	2.43	3.20	yes
B, C	3.63	2.62	3.55	yes
A + B, C	13.67	2.40	3.17	yes

*$F = (N - 2)[(R_1 + R_2 - R)/(N - R_1 - R_2)]$, where R_1 and R_2 are the resultant vectors of each population and R is the resultant vector of the two populations.

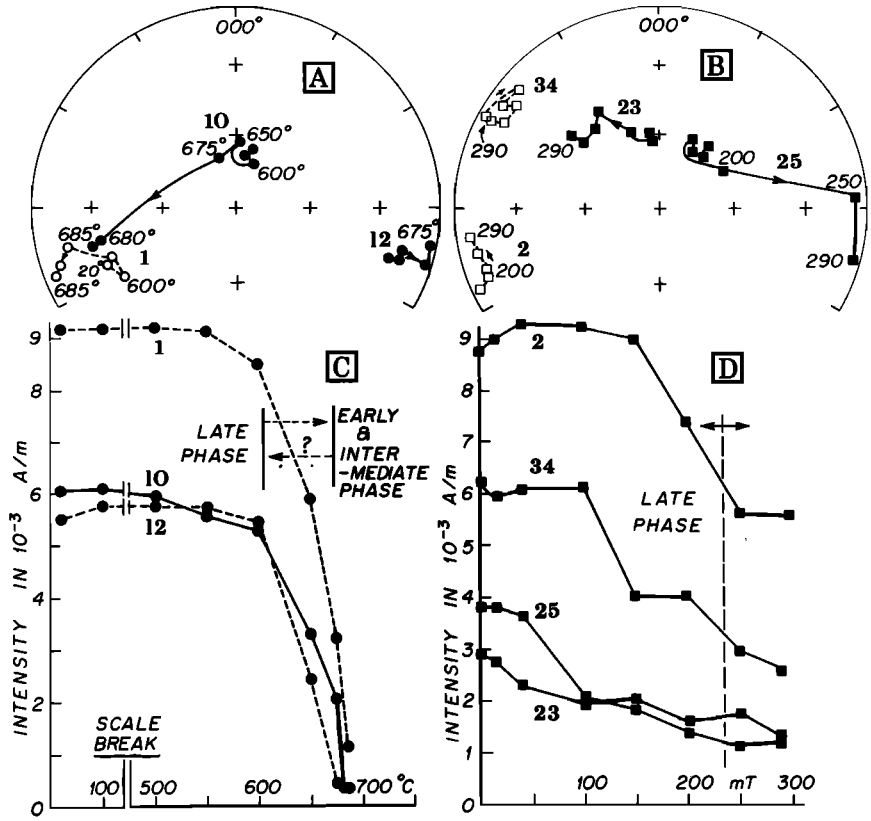


Fig. 11. Comparison of remanence decay curves of (*c*) thermally treated and (*d*) af-treated specimens carrying the steeply inclined J_2 (group B) with those of specimens carrying only J_1 (group A). Group B specimens are identified by steep inclinations (>45°) prior to treatment. Representation of some results has been omitted for reasons given in the the Figure 8 legend. The cleaning steps are shown in Figures 11*c* and 11*d*, plus the 200°C, 300°C, and 400°C treatments which were also performed. The similarity of the magnetic characteristics of the two groups favors interpretation 2 over interpretation 3 (see text). The very high T_B (between 675°C and 680°C) shown in Figure 11*a* indicates that in some specimens the acquisition of J_2 possibly began during the intermediate phase.

plate tectonics to explain some of those isolated poles (see *Irving and McGlynn* [1976] for a review of paths).

The path shows two main features: a single looping into the southern hemisphere between ≃−1100 and −950 m.y. and the existence of a Hadrynian Track which takes the pole (relative to North America) from the equator at about 160°E (≃−950 m.y.) and brings it up over the present day pole and then down to the equator at about 340°E (≃670 m.y.). This path can accommodate all other poles whose ages, although not accurately known, possibly fall in the −1100- to −670-m.y. bracket. However, it should be emphasized that the fitting of a pole to a particular portion of the path does not upgrade the reliability of that pole. Different portions of the Precambrian polar path appear to overlap, so that this method of estimating age of magnetization is hazardous. For example, owing to the proximity of the descending path (Figure 12) and the ascending path (Figure 13), a pole near the equator could belong to either of them.

The path starts north of the equator, with the south trending Keweenaw Track defined by a number of poles from the Keweenawan, including two KRP's; this trend is supported by seven poles (6–12) from a stratigraphic sequence in the Grand Canyon Supergroup [*Elston and Grommé*, 1974]. As was shown previously (Figure 9), the path is now extended southward over the equator with the KRP's Nonesuch (NO), Freda (FR), and Jacobsville $J_1(C) \rightarrow J_1(B) \rightarrow J_1(A)$. This southward extension is supported by the Rapitan Group pole (RX) from the Mackenzie Mountains obtained after extensive chemical, thermal, and af treatments [*Morris*, 1977*a*]. The path has been drawn further southward to include the Grenville KRP's (AL, H(A), M_1, and MG), and this brings us to the Grenville problem, which is now reviewed briefly.

THE GRENVILLE PROBLEM

Several proposals have been advanced to explain the poles obtained from the Grenville Structural Province (G poles). One of the difficulties in the interpretation of the results is that they were obtained from metamorphic terrain and the age of magnetization is uncertain. Radiometric dating of Grenville rocks ranges from −1500 to −950 m.y. [*Wynne-Edwards*, 1976]. When the G poles are compared to the Interior Laurentia (IL being Laurentia without the Grenville Province) for that period of time, they are found to be discordant; that is, most of the G poles lie south of the equator, while most of the IL poles lie north of the equator. Two models could explain this: (1) The two-plate model—the G poles have their time equivalents in the existing IL polar path, and the difference in pole locations indicates that the Grenville Province was once thousands of kilometers away from Interior Laurentia [*Irving et al.*, 1972; *Palmer and Carmichael*, 1973; *Buchan and Dunlop*, 1973, 1976; *Irving et al.*, 1974; *Ueno et al.*, 1975]. (2) The one-plate model—the G poles record a portion of a single polar path (for the whole of Laurentia) which has not yet been observed in Interior Laurentia. The G poles are then incorporated into the Laurentia polar path by means of a loop. Different age assignments have been given to such a loop: −1400 to

−1250 m.y. [*Roy and Fahrig*, 1973; *Fahrig et al.*, 1974], −1100 to −950 m.y. [*McWilliams and Dunlop*, 1975], and −800 to − 600 m.y. [*Stewart and Irving*, 1974; *Irving et al.*, 1974; *Ueno et al.*, 1975].

The reason that so many proposals have been made is that the polar path for Interior Laurentia is not known in detail. Many gaps still exist. The G poles could possibly fall into any of them, and this weakens the case for a two-plate model. Alternately, the weakness of the proposed loops is that they have been constructed from G poles solely. Furthermore, none of the proposals have been able to explain all the G poles.

The loop shown in Figures 12 and 13 is supported by IL poles. The Jacobsville ($J_1(B) \rightarrow J_1(A) \rightarrow J_1(SA)$), Rapitan Group (RX), and Nankoweap (NP) results indicate that the IL polar path executed a loop into the southern hemisphere some time in the −1100- to −950-m.y. period. A definite answer to this question must evidently await a more complete definition of the loop by obtaining more IL data. It should be noted that the tentative placing of the Grenville KRP's on this loop is in accordance with the most probable age of magnetization. Detailed experimental work has shown that the AL [*Hargraves and Burt*, 1967], the H(A) [*Buchan and Dunlop*, 1973, 1976], and the MG [*McWilliams and Dunlop*, 1975] magnetizations are carried by both magnetite and hematite; the M_1 magnetization [*Irving et al.*, 1974] is carried by hematite. It would appear then that (in each instance) the magnetization was acquired during cooling following metamorphism, as was suggested by the respective authors. The best estimate of age is −1100 to −1000 m.y. This loop also accommodates all the Grenville poles, including the M_E pole (Mealy Mountain, east component [*Fahrig et al.*, 1974]), which has remained unexplained in most of the previous interpretations. It is noted that the SH pole [*Fahrig et al.*, 1974] from the Shabogamo Gabbro near the Grenville Front falls between poles $J_1(A)$ and RX; this pole, however, could possibly belong to an earlier part of the Precambrian polar path.

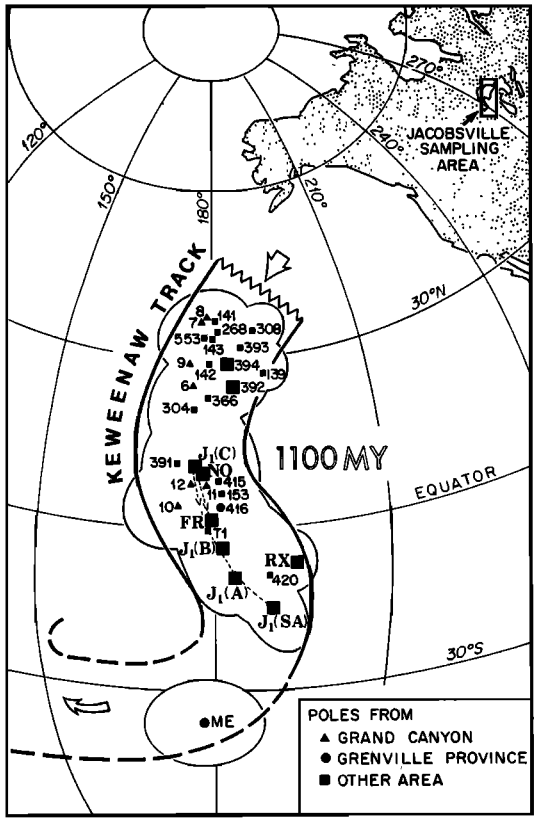


Fig. 12. South trending polar path relative to North America 1100 m.y. ago. For Figures 12, 13, and 14, poles with three digits are listed in the second issue of the Ottawa catalog [*Irving and Hastie*, 1975]; poles represented by triangles (6–12) are from the Grand Canyon Supergroup [*Elston and Grommé*, 1974]; and all other poles identified by letters are referenced in the text. Large symbols represent key reference poles (KRP). Dots represent G poles of rock units within the Grenville Province. Approximate ovals of confidence (P = 0.05) are given individually or contoured where there is a concentration of poles. Solid ovals indicate that the site was taken as unit weight; dashed ovals indicate, that the specimen was taken as unit weight. These ovals should be regarded as a rough estimate only of the accuracy of determination of the pole. The shape and size of an oval is dependent upon the projection used. It should also be noted that the oval axes (δp and δm) for a pole derived from a steep inclination of magnetization can be up to 4 and 2 times larger, respectively, than the oval axes for a pole derived from a low inclination of magnetization. Lambert equal area projection centered on 180°E, 30°N.

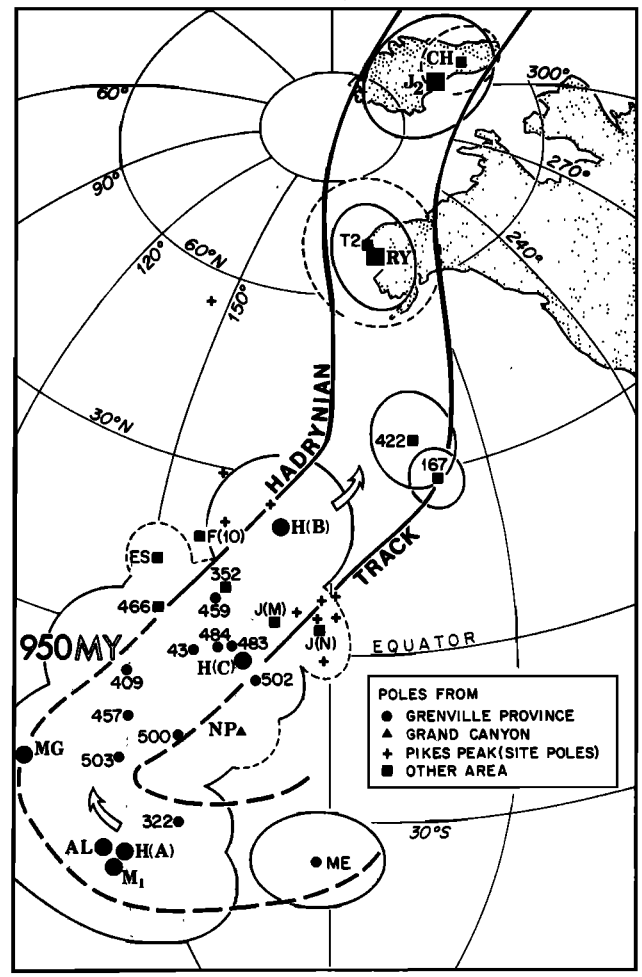


Fig. 13. North trending part of the Hadrynian Track. The Grenville poles can be fitted into the southern hemisphere loop required by the Jacobsville and other poles from Interior Laurentia. Pole identifications are described in the Figure 12 legend. NP is the Nankoweap Formation pole [*Elston and Grommé*, 1974], and ES is the Eileen Sandstone pole [*Dubois*, 1962]. The site poles of the Pikes Peak Granite [*Spall*, 1971] are shown individually. Pole F(10) represents the intermediate phase remanence of 13 in Figure 10*a*. The J(N) pole is derived from a rough estimate of the late phase remanence of the SA section (see text). The J(M) pole is from *French* [1976]. It is estimated that the pole path crossed the equator at about 950 m.y. Lambert equal area projection centered on 180°E, 30°N.

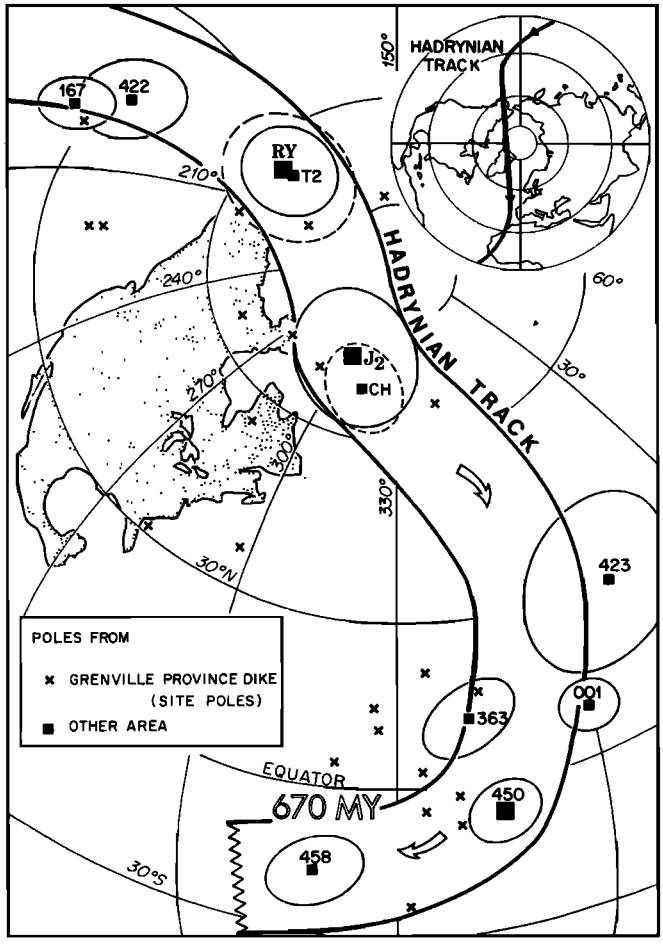


Fig. 14. The Hadrynian Track shown on Lambert equal area projection centered on 330°E, 60°N and geographic north pole in inset. Pole identifications are described in the Figure 12 legend. The site poles of the Grenville Province dikes [*Murthy*, 1971] are shown individually. The Hadrynian Track is not calibrated except for its approximate age limits of −950 to −670 m.y. from equator to equator.

The Hadrynian Track

The tracing of the path into the Hadrynian Track is supported by another study, the sequence H(A) → H(C) →H(B) of the Haliburton intrusives as suggested by *Buchan and Dunlop* [1976]. There are also several indications that the late and intermediate phase remanences of the Jacobsville may be spread along this track. For example, the intermediate phase remanence at site 13 (Figure 10) yields a pole at 157°E, 18°N; the late phase remanence of the SA section (as was mentioned earlier), a pole at 179°E, 04°N; and the thermally obtained remanence of *French* [1976], a pole at 171°E, 05°N.

The KRP's for the Hadrynian Track are J_2 and RY. J_2 has been shown to postdate J_1. Likewise, RY has high T_B and rcf and postdates RY [*Morris*, 1977*a*]. Supporting evidence is provided by the Orienta 2 (T2) and Chequamegon (CH) Keweenaw poles [*Dubois*, 1962]. It is noted that the correspondence in the relationship of J_2, T2, and RY with their respective J_1, T1, and RX counterparts is remarkable. Other possible evidence for high-latitude poles includes some of the secondary site mean poles of the Pikes Peak Granite [*Spall*, 1971] and the site poles from the Grenville Province diabase dikes [*Murthy*, 1971]. These dikes postdate the Grenville metamorphic episode and can be divided into two groups, one being tightly grouped about the Franklin pole (450) and the other yielding poles which roughly follow the Hadrynian Track. The track terminates with the KRP Franklin diabase (450) obtained from several studies and the Tudor Gabbro (TG) [*Palmer and Carmichael*, 1973]. The placing of the latter pole is in accordance with the reported K-Ar isochron age [*Hayatsu and Palmer*, 1975].

Except for its age limits, about −950 to −670 m.y. from equator to equator, the Hadrynian Track cannot be calibrated because it is derived from secondary or late phase magnetizations whose ages are not known with precision with one possible exception. Only one direction of magnetization was found in the Chequamegon Sandstone (CH) [*Dubois*, 1962]. The directions of 15 samples after thermal treatment are tightly grouped ($k = 46$, $\alpha_{95} = 5°$). The mean direction is 029°, +74°, yielding a pole at 308°E, 68°N. Without chemical experiments to separate all phases of magnetization the time of acquisition cannot be determined. Dubois found that the directions remained constant during thermal demagnetization, suggesting, by analogy with the Jacobsville SA results, that the direction of the field may not have changed significantly during the magnetization process. If so, this direction would represent the direction of the field at or shortly after deposition of the Chequamegon Sandstone. 'This sandstone is the uppermost Keweenawan formation known' [*Dubois*, 1962, p. 39]. It then appears that chemical experiments performed on a more extensive collection of the Chequamegon Sandstone might be of prime importance for calibration of the Hadrynian Track.

Conclusions

The necessity of performing chemical leaching experiments on red beds has been clearly demonstrated. Without the aid of this technique, only part of the information contained in these beds would have been recovered. Several studies have now shown that the remanence acquisition process of red beds is lengthy and that (at least) three phases of magnetization can be recognized and successively separated by acid leaching. The experiments indicate a dependence between the resistance to acid leaching of the different magnetic carriers and their time of formation, the carriers formed at the end of the process being the least resistant. The sequence of magnetization can therefore be established, meaning that red beds can be an extremely valuable tool for the construction of polar paths. Thermal and/or af treatments alone are insufficient for the study of red beds, since they cannot recognize and separate all phases of magnetization. Nevertheless, they are useful because they can provide one with the T_B and rcf of certain phases. Because each treatment provides one with additional information, the use of the three treatments is advantageous. Multiphase magnetizations appear to be common in igneous rocks also, and separation of those phases could be used to construct polar paths. However, it is often difficult to establish the sequence of magnetization because the remanence phase with the highest T_B and/or rcf does not necessarily predate the others. It then appears logical to use the results of thoroughly studied rock units as reference points for the construction of polar paths.

Sequences of magnetization and a few well-determined poles derived from detailed laboratory work have been used to sketch an apparent polar path for the period from −1100 to −650 m.y. This path departs greatly from previously proposed paths. Nevertheless, it can accommodate all other poles whose ages have been assumed to fall into that age bracket. In our view, the drawing of more complicated patterns to explain

poles obtained after limited and inadequate cleaning treatments is not justified; it must be ascertained first that all phases of magnetization have been at least recognized and if possible separated.

Comparison of the J_1 poles with other Keweenawan poles and the trend of the path indicate that the Jacobsville is most probably of upper Keweenawan age ($\simeq -1100$ m.y.). A distinction should, however, be made between the sediments of Sault Ste. Marie and those of the Keweenaw Peninsula. The former appear to have been deposited contemporaneously with the Nonesuch-Freda of Michigan. The Jacobsville on the Keweenaw Peninsula is somewhat younger. Its exact age cannot be determined because of scarcity of data. The agreement between the Chequamegon pole (CH) and the J_2 pole which marks the end of the remanence acquisition process of the Jacobsville indicates that the Jacobsville is not younger (and probably older) than the Chequamegon. The similarity between the Orienta T1 pole and the Jacobsville $J_1(A)$ pole indicates that the two formations might be of comparable age. Chemical leaching experiments performed on the Chequamegon and the Orienta should provide one with the relative ages of those three formations.

The large apparent polar movement (180°) that took place during the Hadrynian should greatly facilitate the stratigraphic control and correlation of rock units of that age. The average pole displacement along the Hadrynian Track is 0.6° latitudinal change/m.y., so that a 20- to 30-m.y. difference (12°–18°) should easily be discernible paleomagnetically. A more precise definition and calibration of the Hadrynian Track is needed, and this might be possible with detailed experimental work of extensive stratigraphic sequences of red beds such as those found in the Chuar Group of the Grand Canyon Supergroup.

This is only the second high-latitude polar path that has been found in the Precambrian, the other having an age of −2300 to −2400 m.y. (see *Morris* [1977*b*] and *Irving and Naldrett* [1977] for review). More probably have occurred; however, it is debatable whether all of them will be discovered. Considering that the areas north of 60°N and south of 60°S account for only 14% of the earth's surface, the likelihood of finding a high-latitude pole is small. This might be reduced even further, owing to the possibility that a magnetization yielding a high-latitude pole might be wrongly attributed to remagnetization in the present field. Therefore the prospect of finding poles in sufficient number to construct a dependable high-latitude polar path appears dim.

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